

Problem Statement

Design Database Schema for a system like Netflix with following Use Cases.

Use Cases

- Netflix has users.
- Every user has an email and a password.
- Users can create profiles to have separate independent environments.
- Each profile has a name and a type. Type can be KID or ADULT.
- There are multiple videos on netflix.
- For each video, there will be a title, description and a cast.
- A cast is a list of actors who were a part of the video. For each actor we need to know their name and list of videos they were a part of.
- For every video, for any profile who watched that video, we need to know the status (COMPLETED/ IN PROGRESS).
- For every profile for whom a video is in progress, we want to know their last watch timestamp.

Entities:

1. Users
id, email, password -> {id}
2. Profiles
id, name, profile_type_id, user_id(fk) -> {id}
3. Videos
id, title, description -> {id}
4. Actors
id, name -> {id}
5. Profile_Types
id, value -> {id}
6. Watch_Video_Status_Type
id, value -> {id}
7. Video_Actors
video_id, actor_id -> {video_id, actor_id}
8. Video_Profiles
video_id, profile_id, video_status_type_id, last_watched_at -> {profile_id, video_id}, Queries on profile_id needs to be faster
9. Video_Status_Types
id, value -> {id}

Relation among entities:

1. Users and Profiles (1:m)
1 m
1 1

2. Videos and Actors (m:m)

1	m
m	1

3. Videos and Profiles (m:m)

1	m
m	1

Schema Design:

1. Identify the tables
2. Add the PK
3. Add other attributes
4. Represent cardinalities for relationships among entities